

Ex. 2

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```
print(uuid::UUIDgenerate())
```

```
## [1] "98474c24-ac0c-42a8-809f-3e6d6793242b"
```

```
# Data
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)

# 1. Range and Class Width
min_val <- min(commute)
max_val <- max(commute)
breaks <- seq(from = min_val, to = max_val, length.out = 6)

# 2. Grouped Frequency
classes <- cut(commute, breaks = breaks, include.lowest = TRUE)
freq_table <- table(classes)

# 3. Cumulative Frequency
cum_freq <- cumsum(freq_table)

# Output Table
freq_df <- data.frame(
  Class_Interval = names(freq_table),
  Frequency = as.vector(freq_table),
  Cumulative_Frequency = cum_freq
)

print(freq_df)
```

```
##           Class_Interval Frequency Cumulative_Frequency
## [18,23.4]      [18,23.4]         5                     5
## (23.4,28.8]    (23.4,28.8]         5                    10
## (28.8,34.2]    (28.8,34.2]         5                    15
## (34.2,39.6]    (34.2,39.6]         5                    20
## (39.6,45]      (39.6,45]          4                    24
```

R Markdown

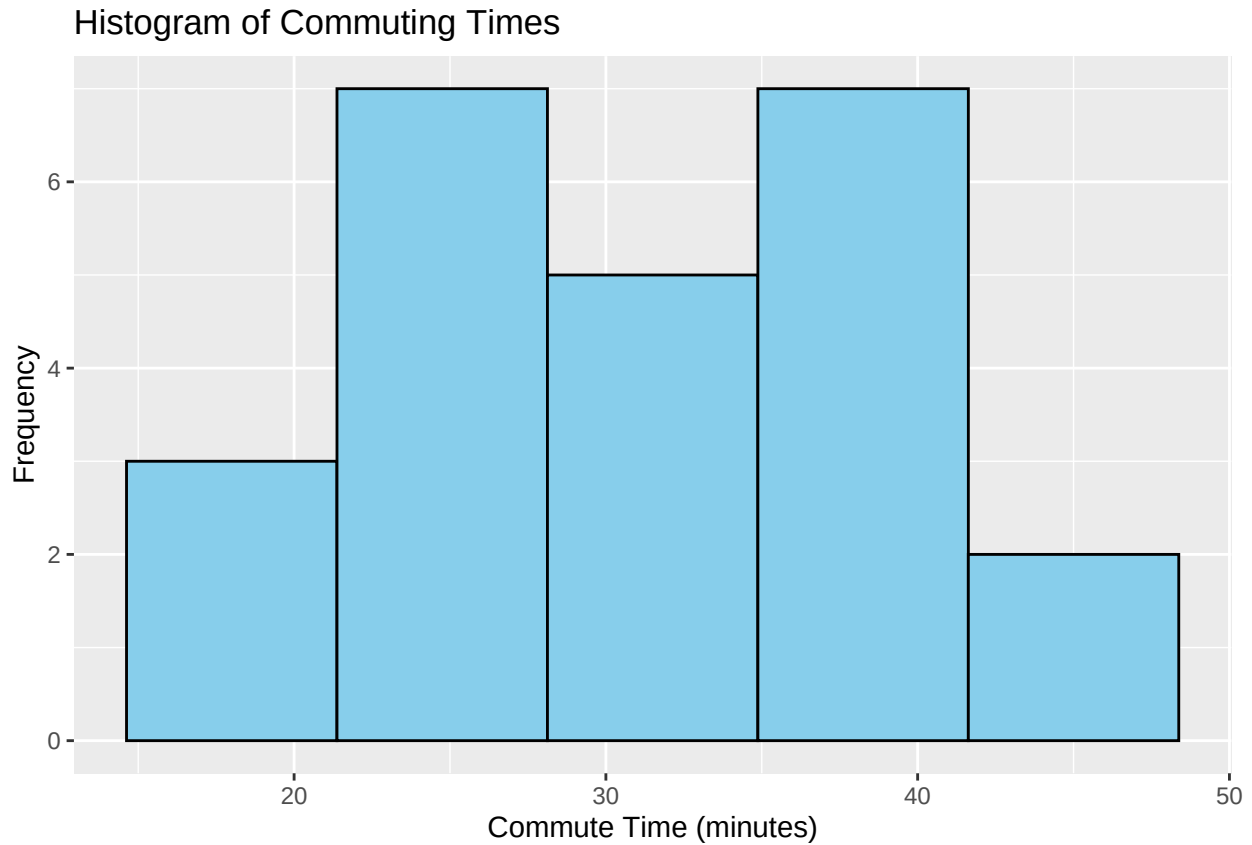
This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
library(ggplot2)

commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)

# Question 2: Histogram
ggplot(data.frame(commute), aes(x = commute)) +
  geom_histogram(bins = 5, color = "black", fill = "skyblue") +
  labs(
    title = "Histogram of Commuting Times",
    x = "Commute Time (minutes)",
    y = "Frequency"
  )
```



```
# Question 3: Ogive
min_val <- min(commute)
max_val <- max(commute)
breaks <- seq(from = min_val, to = max_val, length.out = 6)
classes <- cut(commute, breaks = breaks, include.lowest = TRUE)
cum_freq <- cumsum(table(classes))

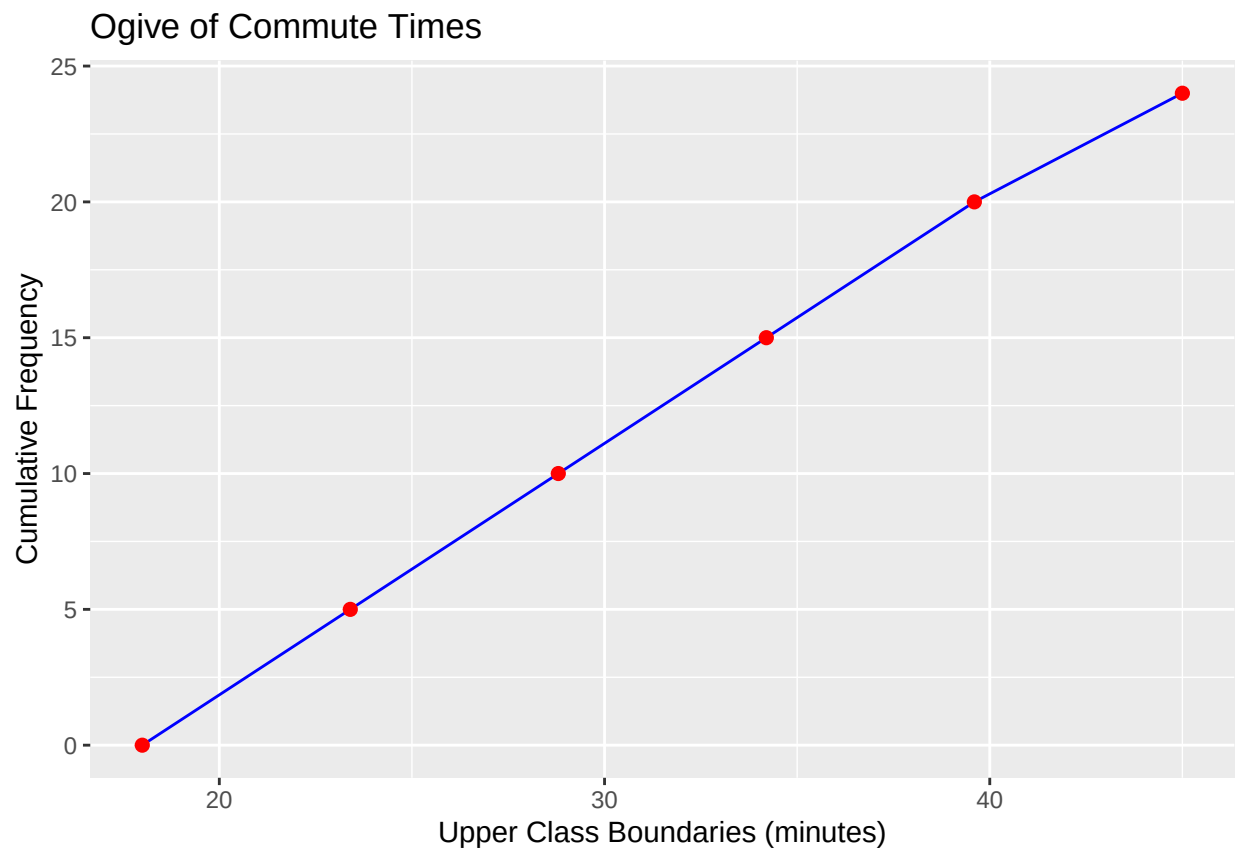
ogive_data <- data.frame(
  x = c(min_val, breaks[-1]),
```

```

  y = c(0, cum_freq)
)

ggplot(ogive_data, aes(x = x, y = y)) +
  geom_line(color = "blue") +
  geom_point(color = "red", size = 2) +
  labs(
    title = "Ogive of Commute Times",
    x = "Upper Class Boundaries (minutes)",
    y = "Cumulative Frequency"
  )
)

```



```

# Estimate students commute less than 35 minutes
sum(commute < 35)

```

```
## [1] 15
```

Including Plots

You can also embed plots, for example:

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.